Dataset & Research Objective

Proposed title: Regime-Aware Fuzzy-Gated Mixture of LSTM Experts for UOB Stock Return Forecasting

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Research question

- Can a regime-aware fuzzy gate dynamically combine LSTM experts to improve next-day stock-return forecasts and simulated trading performance?

Financial dataset

Asset / symbol	UOB Group / U11.SI (Singapore-listed bank)
Source & frequency	Yahoo Finance; daily historical OHLCV prices
Study period	January 2015–September 2026 (planned; confirm download coverage)
Target	Next-trading-day log return: ln(C[t+1] / C[t])
Input window	Previous 30 trading days of price-derived features

Features and preprocessing

- Use price/volume series plus engineered returns and price trends; calculate 20-day rolling volatility and 50-day trend slope.
- Use the Straits Times Index (STI) drawdown relative to its previous one-year high as a wider market-regime indicator.
- Check missing values, corporate actions and price-adjustment consistency. Split data chronologically; fit normalization and fuzzy-set parameters using training data only.
- All inputs and derived features available to a prediction must be constructed solely from information observed by that prediction time.

Primary hypothesis

- Market regimes (e.g., calm versus stressed, upward versus downward trends) may require different forecasting behaviours. A fuzzy gate can learn varying weights for specialist LSTMs.

UOB historical data: https://sg.finance.yahoo.com/quote/U11.SI/history/

STI reference series: https://sg.finance.yahoo.com/quote/%5ESTI/history/

Hybrid AI Method: TSK Fuzzy-Gated LSTM Experts

Core innovation: A Takagi–Sugeno–Kang (TSK)-style fuzzy gate combines three small LSTM experts, instead of relying on one fixed LSTM for all market regimes.

Proposed architecture

Component	Role
Three LSTM experts	Each receives a 30-day financial feature sequence and predicts next-day log return.
Regime inputs	20-day volatility; 50-day trend slope; STI drawdown from prior one-year high.
Fuzzy gate	Trainable Gaussian membership functions express regime states (low / medium / high) and derive rule-based expert weights.
Fusion / output	Weighted sum of the three experts' forecasts produces the final next-day log-return prediction.

Illustrative interpretable rule

"IF volatility is high AND the trend is downward, assign a greater weight to Expert 3." This is a proposed rule pattern, not a verified learned result.

Training and analysis

- Train expert networks and fuzzy-gating parameters using a regression loss; tune hyperparameters only on validation periods.
- Address possible expert collapse (one expert receiving almost all weight) with monitoring and, if needed, a mild balancing regularizer.
- Plot membership functions and expert weights across calm and stressed periods, especially 2020 and 2022; test whether weight changes meaningfully track market regimes.
- Implement this proposed gate-and-experts extension ourselves. The existing Fuzzy LSTM repository is a reference/baseline, not confirmed to contain this exact architecture.

Research/code starting points

Original Fuzzy LSTM reference notebooks: https://github.com/MakhovaAnastasia/Paper_Fuzzy_LSTM

Related fuzzy recurrent-network publication: https://doi.org/10.1007/978-3-032-15761-4_4

Note: Method details and literature attribution will be refined after reviewing the papers and reproducing the reference code.

Benchmark, Evaluation & Open Resources

Benchmarks and ablations

- Primary: Standard LSTM and the existing fuzzy-feature LSTM, each trained on the SAME UOB target, horizon and chronological split.
- Fair-capacity control: one LSTM with approximately the same total trainable parameter count as the three-expert model.
- Optional additional baselines if feasible: naïve forecast, DLinear or ANFIS. Buy-and-hold is a trading benchmark, not a return-forecasting model.

Evaluation methodology

- Use chronological training, validation and held-out test periods; conduct walk-forward tests without random shuffling or future-data leakage.
- Proposed robust protocol: historical annual test folds (2020–2024), validation at the end of each earlier training segment, and January 2025–September 2026 as a final held-out period; confirm fold feasibility.
- Report RMSE and MAE on log returns, directional accuracy, and (when statistically justified) repeated-seed variability.
- Backtest a consistent long-or-cash strategy with signals formed after information becomes available, next-tradable-price execution, realistic transaction costs, and comparison against buy-and-hold.
- Report net cumulative return, Sharpe ratio, maximum drawdown and turnover. Include regime-wise performance and expert-weight visualizations.

Optional extension — not required for the initial proposal

- Investigate transfer learning by pre-training on DBS and OCBC and fine-tuning on UOB if time permits.

Open resources (full links)

UOB / Yahoo Finance: https://sg.finance.yahoo.com/quote/U11.SI/history/

DBS / Yahoo Finance (optional transfer): https://sg.finance.yahoo.com/quote/D05.SI/history/

OCBC / Yahoo Finance (optional transfer): https://sg.finance.yahoo.com/quote/O39.SI/history/

Fuzzy LSTM reference code: https://github.com/MakhovaAnastasia/Paper_Fuzzy_LSTM

Related publication DOI: https://doi.org/10.1007/978-3-032-15761-4_4

Course project sample Colab: https://colab.research.google.com/drive/14VEWf5KzSxrTm8V4Px6cDu3eN3KUJXk5

Project submission: proposal is a planned method, not a claim of completed experiments; final design remains subject to instructor approval.